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Inference and Applications of Ancestral Recombination Graphs

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Abstract

Ancestral Recombination Graphs (ARGs) are graphs that summarize all the complex genealogical relationships among individuals represented in a sample of DNA sequences. Their use is currently revolutionizing the field of population genetics and is leading to the development of powerful new methods for elucidating individual and population genetic processes, including population size history, migration, admixture, recombination, mutation, and selection. In this review we will introduce the readers to the structure of ARGs and discuss how they relate to processes such as recombination and genetic drift. We will explore differences and similargeities between methods for estimating ARGs and provide concrete illustrative examples of how ARGs can be used to elucidate population level processes.

Keywords

Coalescence Tree; Ancestral Recombination Graph; Population Genetics; Recombination

1 Introduction

Coalescent theory has formed the foundation for analyses of population genetic data since the invention of PCR and DNA sequencing technologies [1-5]. The central object of coalescent theory is the coalescence tree, which summarizes information about the genetic relationships among sampled individuals. The leaves (tips) of the tree represent individual DNA sequences in the sample, the edges (lineages) represent lines of descent, and the internal nodes represent coalescence events, i.e. the times at which specific individuals, or groups of individuals, in the sample find their most recent common ancestor (MRCA). The root of the tree represents the MRCA of all individuals represented in the sample. An example of a coalescence tree is given in Figure 1A.

Kingman [1] and Hudson [2, 6] discovered in the early 1980s that the relationship among individuals in the sample could be described by a binary tree, i.e. a tree in which each node has either 0 children (leaf nodes) or two children (internal nodes), and that population

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genetic models of genetic drift provide simple predictions regarding the structure of the tree. This quickly led to the emergence of many statistical methods for inferring population level parameters such as population size changes [7] and migration rates [8-10], and it spurred the development of new methods for detecting natural selection using population genetic data [6, 11, 12]. The central insight forming the basis for these methods is that all the information in DNA sequencing data is represented by the coalescent tree. If the coalescent tree can be inferred with 100% accuracy, there would be no more information to obtain about these demographic processes from the sequence data. This principle led to the development of full likelihood and Bayesian statistical methods that achieved their objectives by integrating over the set of possible coalescence trees using Markov Chain Monte Carlo (MCMC) or other stochastic methods, i.e. they took uncertainty regarding the structure of the coalescent trees into account by considering many trees weighted by their relative probabilities [9, 10, 13, 14]. By the early 2000s, a mature set of tree-based statistical methods for analyzing DNA sequencing data had emerged. There was only one problem: these methods assumed no recombination and were therefore only really useful for mtDNA, chloroplast DNA, some viral sequencing data, and a few other types of markers that are not subject to recombination. However, with the emergence of next-generation sequencing technologies, these methods were rapidly becoming obsolete as the focus now changed to genomic data with an abundance of recombination. For genomic data, there is not just one coalescent tree, but as many coalescent trees as there are recombination events in the history of the data, since each recombination event changes the tree from one to another. Some approaches attempted to deal with this problem by assuming that the genome could be divided into short genomic regions with free recombination between regions but no recombination within regions [15, 16]. However, the mutation rate is on the same order of magnitude as the recombination rate for many eukaryotes including humans [17, 18], implying that every time a mutation event occurs that can provide information about the coalescence tree, a recombination event will in average also occur, obscuring the information regarding the tree structure. Furthermore, many of the methods aimed at short non-recombining regions simply did not scale up computationally to thousands of genomic regions. Consequently, the field moved slowly away from full likelihood and Bayesian methods and concentrated instead on composite likelihood methods based on treating SNPs as if they were independent [19-21], focusing solely on the distribution of allele frequencies [22-24], using Approximate Bayesian Computation (ABC) methods [25], or using other methods that either only used a small subset of the information in the data or only focused on pairs of sequences [26].

This was the state of the field until 2014, when a disruptive paper by Rasmussen and colleagues demonstrated that inferences of coalescent trees in models with recombination, in fact, is computational feasible, even for genomic scale data [27]. The authors showed, using clever sampling methods, that it is possible to estimate (and sample from a posterior) the coalescent trees along the length of the genome in a joint structure called an Ancestral Recombination Graph (ARG). This opened up the possibility of extracting much more of the information arising from population genetic sequencing data and, perhaps, developing full-likelihood and Bayesian methods for inference in population genetics using ARGs in place of individual coalescence trees.

2 What are Ancestral Recombination Graphs (ARGs)?

The coalescent process with recombination was first described by Hudson [2]. In later work Griffiths and Marjoram [28] developed this process further and also coined the term “Ancestral Recombination Graph” as the random graph structure arising as a product of the coalescent with recombination (abbreviated as CwR) originally described by Hudson [2]. The basic idea of ARGs is that they represent a coalescent process in which ancestral lineages may not only merge by coalescence, but also split due to recombination. An example of an ARG is shown in Figure 1D. There are three sequences, and the ARG traces the coalescent process back in time until an MRCA has been found. Each recombination event (1 and 2 in Figure 1D) breaks up the DNA sequence into two segments. The coalescence events (3, 4, 5, and 6 in Figure 1D) merge lineages together and represent the MRCA of these lineages. In Figure 1D, the DNA segments containing ancestral genetic material for the three DNA sequences, are depicted adjacent to each edge. For example, the lineage going from node 2 to node 4 only contains ancestral material for the first green segment of the DNA sequence. The rest of the DNA sequence on this lineage is not ancestral to any of the three individuals in the sample.

From the ARG, the individual coalescence trees for each segment of the genome can be deduced. Figure 1(A-C) shows the coalescent trees for the green, blue, and red segments of the DNA sequence respectively. Notice how these trees in Figure 1(A-C) can be deduced from Figure 1D. Recombination will often generate trees with different topologies, although sometimes they may just differ in having different branch lengths, as in the example in Figure 1. The ARG contains the information regarding all the coalescent trees in the genome, but it also contains information about recombination events and their location on the lineages of the coalescent trees. Clearly, for genomic data with many individuals and tens or hundreds of thousands of recombination events, the ARG can become very complicated. As a result, it is common to make some simplifications. In particular, the time of a recombination event on a lineage cannot be inferred from data and is, therefore, often not represented in the ARG. Further simplifications and approximations are often used to make inferences based on ARGs feasible.

The stochastic process generating ARGs, as described by Hudson [2] and Griffiths and Marjoram [28], is a process running from the present backwards in time, allowing recombination events and coalescence events to occur until an MRCA has been found for each locus. Another way of looking at the process generating ARGs is to consider how coalescent trees change along the length of the genome. As the trees can be deduced from the ARG, it should be possible to define a process that runs from one end of the chromosome to the other end, generating coalescent trees in the process. Wiuf and Hein [29] considered this problem and showed that this process was non-Markovian, meaning that the probability distribution of a coalescent tree in one position does not only depend on the coalescent tree in the previous position, but also depends on all the other trees along the length of the chromosome. This makes considering the sequential process of trees, as they change along the length of the chromosome, less useful in applications because it is necessary simultaneously to keep track of all trees for purposes such as inference and simulation. To make a simpler and more computationally tractable process, McVean and

Cardin [30] proposed a model that could approximate the tree-generating process along the length of the genome as a Markovian process which they dubbed the Sequentially Markov Coalescent (SMC). The idea in that process was to disallow coalescence events between edges that do not share any genetic material. For example, in Figure 1D a coalescence event happens between two lineages, which only contain ancestral genetic material for the green and the red genomic segments respectively, forming node 4. Under the SMC process, such coalescence events are not allowed to happen, leading to a much-simplified process. A modification of the SMC process, called SMC′, was proposed by Wall and Marjoram [31]. While the SMC process only allows coalescence events between lineages that contain at least some shared ancestral genetic material, SMC′ also allows coalescence events between lineages with no shared genetic material, but with genetic material from adjacent loci. The possible coalescence events allowed under each of these three models is detailed in Figure 2. This small extension of the SMC does not lead to much added computational complexity, but it drastically improves the accuracy of the approximation. Subsequent work [32] has shown that, by most measures, the SMC′ process is a very close approximation to the full coalescence process with recombination as described by Griffiths and Marjoram [28] and is today, arguably, the preferred process for population genetic modeling of the coalescent process with recombination.

Different assumptions about the process generating ARGs lead to different definitions of ARGs (see, for example, [33]). Some ARGs allowed in the full coalescent process with recombination are not supported by the SMC′ process, and further ARGs that are supported by the SMC′ process are not allowed by the SMC process. Sometimes, ARGs can be represented even more simply, as just a sequence of trees without explicit information about recombination and/or identification of which coalescence events are shared among trees [34, 35]. It is then a matter of definition if these sequences of trees are truly ARGs. However, no matter how ARGs are defined, they can be considered as products of a coalescent process governed by population genetic factors such as population sizes, migration rates, and selection, and molecular forces such as mutation and recombination. The ARGs contain information about these processes, and methods based on ARGs that can extract this information have the potential to take on a central role in population genetics in the genomic era, similar to that of coalescence trees in the 1990s.

3 Inference of ARGs

3.1 Paradigms of ARG inference

Many different methods for inferring ARGs have been developed, including tsinfer+tsdate [36, 37], ARG-Needle [38], SINGER [39] and ARGweaver [27]. These methods differ in many ways, including the way they represent ARGs., which is connected to implicit assumptions regarding the underlying generative process. Methods representing ARGs as a series of coalescent trees, without necessarily enforcing that the trees share nodes or branches with each other include Relate [40] and Margarita [34]). This representation is at times convenient but is not efficient in that many trees embedded in ARGs share the same nodes and edges (as described in Figure 1). Furthermore, this representation destroys some of the naturally occurring correlation structure between trees. As a result, a more compact representation is sometimes used where nodes and edges are only represented once even though they may be components of many trees. This type of representation underlies simulation algorithms such as FastCoal [31], fastsimcoal/fastsimcoal2 [41, 42], and msprime [43, 44], as well as inference programs such as tsinfer [36], ARG-Needle [38], SINGER [39] and ARGweaver [27]. It can also lead to a compact data structure for the sequence data, as each mutation only needs to be represented once, even if it appears in a large number of individuals [45]. The representation of an ARG as a sequence of correlated trees with shared nodes is sometimes referred to as a “tree sequence” (see [43, 46] for further discussion of this concept).

The inference of ARGs is a computationally difficult problem because the number of possible graph topologies can be enormous. Additionally, the observed sequence data provides limited information about the structure of the local tree in any particular position of the genome. For example, the mutation and recombination rates in humans are fairly similar, meaning that, on average, every time a new mutation occurs that might provide information about the tree structure, a new recombination event will also occur that changes the tree structure. Some methods (e.g. ARGweaver and SINGER) address this problem by taking a Bayesian approach, where ARGs are sampled from a posterior distribution. Instead of inferring a single ARG, many possible ARGs are sampled, which provides a representation of the uncertainty in ARG inference and facilitates rigorous downstream statistical analyses. Other methods, such as Relate, tsinfer, and ARG-Needle, only infer a single topology, although Relate additionally samples coalescence times for the inferred topology. As the sampling of ARGs is computationally demanding, methods that only estimate a single ARG tend to be much faster than methods that sample multiple ARGs.

3.2 Current methods for ARG inference

There have been several methods developed for ARG inference [34, 37, 39, 40, 47-53], and in this section we provide an overview of some of them.

ARGweaver is a Bayesian MCMC method that produces samples of ARGs from a posterior, thereby quantifying the uncertainty in ARG estimation. As with all Bayesian methods, a prior is required. In the original implementation, a coalescence prior assuming a single population of constant size was used. However, a recent update of the program, ARGweaver-D, can incorporate more complicated priors such as population splits and migrations [54]. Algorithmically, ARGweaver [47] is based on an approach called “threading”. In threading, a new DNA sequence is fitted into an existing ARG by sequentially determining recombination points and coalescence events along its genomic sequence. ARGweaver starts with just two sequences and then iteratively threads more sequences to an expanding ARG until all sequences have been threaded. It then repeatedly re-threads sequences in such a way that the resulting distribution of ARGs represents the correct posterior distribution.

The method Relate [40, 55] is based on estimating topologies for different genomic segments independently using pairwise distance matrices. The pairwise distance matrices are obtained using an approximative model of sequence similarity first described by Li and Stephens in 2003 [56]. After the topologies have been estimated, mutations are mapped to branches and branch lengths are estimated using a Markov Chain Monte Carlo (MCMC) algorithm. In this way, while a single topology is assumed for each segment, uncertainty about the branch lengths of each topology is represented in the MCMC output.

The method tsinfer [36] first infers ancestral haplotypes from patterns of mutations shared along the genomes, with ages assigned to them based on their frequencies. Then, coalescence events in the ARG are inferred using the Li and Stephens model [56] by comparing each ancestral haplotype to more ancient ones. After the topology has been estimated, another algorithm (tsdate [37]), can be used to estimate the branch lengths based on the distribution of mutations on branches and a coalescent prior.

ARG-Needle [38], like ARGweaver, is based on the concept of "threading". It performs threading very quickly by finding, for different stretches of the genome, closely related leaf nodes already in the graph using sequence identity. It then uses a coalescent Hidden Markov Model (HMM) [57] to estimate the times at which the new sequence coalesces with these candidate leaf nodes. The leaf node with the lowest local coalescence time is chosen, and the sequence being threaded then joins the ARG at the ancestral lineages of the chosen leaf nodes at the times inferred from the HMM.

SINGER [39] is conceptually similar to ARGweaver in that it provides Bayesian estimates of the distribution of ARGs by using threading in an MCMC algorithm. However, it differs from ARGweaver in the details of the algorithm. For example, it approximates the threading process by first threading the new sequence with respect to topology and then subsequently assigning coalescence times, which reduces the computational complexity.

These methods also differ in multiple other ways. For example, ARGweaver-D, Relate, and tsinfer+tsdate can incorporate non-contemporary samples, while SINGER and ARG-Needle cannot. ARG-Needle is the only method which explicitly supports sparse genotype-array data. In terms of scalability, ARGweaver-D and SINGER cannot yet be used for 1000 Genomes Project scale data, and only tsinfer+tsdate and ARG-Needle can handle UK Biobank scale data. The applicability of these different methods is summarized in Table 1.

3.3 Accuracy and computational speed

The aforementioned methods have different runtime and accuracy performances. In terms of scalability, tsinfer and ARG-Needle are the most scalable methods, capable of handling hundreds of thousands of sequences [36, 38]. Relate is scalable to a few thousand of sequences, SINGER can handle a few hundred sequences, and ARGweaver can handle dozens of sequences [27, 39]. Simulation studies have found that the fastest methods, such as tsinfer+tsdate, in general have the worst performance and that the more recent methods tend to perform better than older methods (e.g., SINGER performing better than ARGweaver)[39]. To illustrate this, we show performance statistics of the inference of mutation age, coalescence times, and tree topologies from simulations using msprime [43] with these parameters: $N_e = 10,000$, $\mu = 2 \times 10^{-8}$, $r = 2 \times 10^{-8}$, with 50 and 300 sequences. As ARGweaver is computationally expensive and ARG-Needle only runs for at least 300 sequences, we only ran ARGweaver for 50 sequences and only ran ARG-Needle for 300 sequences. We compare methods in terms of mean squared error (MSE) and correlation between the truth and the estimated value of estimates of allele age (Figure 3A) and pairiwse (TMRCA) (Figure 3B). Methods with a low MSE and high correlation performs best. As it is difficult to retrieve allele ages from ARGweaver output and allele ages are not estimated in ARG-Needle, we only show benchmarks for allele age estimation for SINGER, Relate and tsinfer+tsdate. We also compare the accuracy of the methods for topology inference measured as the proportion of rooted subtrees with three, or four, leaf nodes that are incorrectly inferred (triplet topology error rate and quartet topology error rate, respectively). In general, the pattern observed in previous studies are recapitulated with SINGER having the best accuracy and tsinfer+tsdate peforming worst.

The code to reproduce these benchmarking results can be found at: https://github.com/YunDeng98/ARG_review.git.

Lastly, we highlight that while we in this section focused on five methods for ARG-inference, many other methods for ARG inference exist [48, 49, 52, 58, 59], using different models or heuristics for estimating ARGs. See [60] for a further discussion of the relative advantages and limitations of these different methods.

4 Application of ARGs for population genetic inferences

4.1 Inference of demography and selection

ARGs can be used to make detailed and powerful inferences regarding population genetic processes, such as population size history, migration, natural selection, mutation rate, and recombination rate.

A major area where ARGs have been of use is in the inference of demography and the genealogical relationship among individuals. The rate of coalescence at a certain time point is inversely proportional to the effective population size (N_e) at that time. Therefore, one can use the temporal density of coalescence times of an ARG to estimate N_e through time. Figure 4A shows how changes in historic population size are reflected in the ARG. The left panel shows an ARG in the presence of constant population size through time. The right panel shows an ARG in the presence of a historic population bottleneck. The reduced N_e during this period results in a higher density of coalescences than in the constant population size history. ARGs have also been used to estimate the parameters of complex demographic models by considering the different possible embeddings of coalescent trees across the genome in the model [61] This idea of considering the possible paths taken by lineages through a demographic model has also been leveraged for local ancestry inference, where each tree along the genome provides information about the path that tree took through the demographic model and, therefore, about the local ancestry at the segment spanned by this tree [62, 63].

ARGs can also be used to infer natural selection. Inference methods for estimating selection coefficients for a single SNP using ARGs have proceeded by extracting the local tree around the focal SNP and then examining the relative coalescent rate of lineages carrying the derived allele and those carrying the ancestral allele. For example, an allele under positive selection ($s > 0$) is expected to increase in frequency forwardin-time more rapidly than a neutral allele ($s = 0$). Therefore, looking backwards in time, the positively selected allele will appear to rapidly decrease in frequency, meaning the number of lineages carrying this allele will quickly become small. We would then expect a high density of coalescence events of lineages with the selected allele, analogous to the previous discussion of small population sizes generating high densities of coalescence events. Figure 4B illustrates this concept. Both panels shows a coalescence tree for a particular region of the genome (marked in gray). A mutation on a lineage is shown as a blue diamond, causing all descendant lineages to inherit this blue allele. If this allele is neutral ($s = 0$; left panel), no difference in coalescence density is expected. However, if this allele is advantageous ($s > 0$; right panel), an increase in the coalescence density of blue lineages is expected. See [64] and [65] for further discussion of this concept. The usage of the coalescence density of lineages carrying different alleles to infer selection can either be done analytically by explicitly computing the likelihood and integrating over the derived allele frequency [66, 67] or by applying machine learning techniques on the inferred ARGs themselves [68, 69]. These methods can also be slightly modified to examine polygenic selection acting on multiple SNPs concurrently [70] or to reconstruct polygenic scores through time [71].

In addition to inferring demography and selection, there have been other exciting applications of ARGs for inferences. For example, Zhang et al. [53] combined inferred ARGs with a linear mixed model framework to refine association analyses of variants with complex traits [53]. A particular strength of their method was the ability to accurately detect large-effect associations of rare variants on sparse genotype array data. Another promising area of research is the usage of ARGs to explicitly model the spatial location of ancestral lineages of a given sample [72-74]. Furthermore, ARGs have been used to study the evolution of the human mutation spectrum by examining the mappings of mutations to the branches of the ARG to study changes in mutation frequency [75] over time. A similar approach of mapping mutations to branches of reconstructed coalescent trees was taken by [76] in order to estimate the ages of alleles, which in turn were used by [77] to infer historic generation times in different human populations. Although controversies exist about the specific methodology used (see [78] for further discussion), the framework used nevertheless illustrates the exciting potential of ARGs to help understand complex molecular and demographic processes. Nascent ARG-based methods have also recently been developed to infer IBD segments [79], detect chromosomal inversions [80], compute time-stratified, tree-based f-statistics [81], simulate and analyze quantitative traits [82, 83], model LD [84], and simulate local ancestries [85].

4.2 Using ARGs for full probabilistic inferences

Although there are always many possible ARGs that are compatible with a given dataset, analyses that use ARGs often rely on a single ARG estimated from the data. This treatment of a single inferred graph as the true graph greatly simplifies subsequent analysis, however, it presents two distinct problems. First, it does not account for uncertainty in the underlying ARG, inducing false certainty in the results of any analysis that uses a single ARG as the true ARG. Second, as this ARG was likely inferred under a certain prior (e.g. no selection, constant population size, no population structure), the inferred ARG will be biased if the true population genetic parameters differ from the specified prior. Therefore, the unobserved nature of the ARG necessitates both an integration over uncertainty in the ARG and a bias-correction due to incorrect prior specification in order to achieve full probabilistic inference. Future methods can leverage ARGs for full probabilistic inferences in one of two ways. The first way is through a statistical technique known as *importance sampling*. In importance sampling, a set of ARGs is sampled under a specified prior. These ARGs are then assigned weights according to their probabilities under another prior. The insight is that by reweighting the samples appropriately, one can approximate the desired posterior distribution of ARGs under any prior and use this distribution to obtain an unbiased estimate of the parameter of interest. We provide an illustration of this framework on an example dataset in Figure 5. First, a large number of ARGs are sampled from the posterior distribution (which uses both the likelihood based on the sequence data and a specified prior). For simplicity of presentation, only 3 sampled graphs are shown. Then, a weight is computed for each sampled graph. ARGs that are more likely under the prior used to generate the ARGs are given low weights, while ARGs that are more likely under the desired prior are given high weights. Finally, the likelihood of a certain parameter value can be approximated as the weighted average of the likelihood of each of the sampled graphs. This approximation becomes exact as more and more samples of ARGs are used.

The alternative approach to importance sampling is to incorporate parameter estimation directly into the MCMC algorithms used to sample ARGs. For example, in addition to proposing changes in ARG topologies and branch lengths, an ARG-sampling algorithm could also propose changes in historic population size or mutation rate. There has been previous work on inferring ARGs under user-defined demographic models. [51] and Relate [40] has the capability to alternate between the estimation of coalescence times and the estimation of historic population sizes, but a true method that jointly samples ARGs and parameters has not yet been developed. This is likely due to the dramatically increased computational cost of doing this joint sampling in addition to the necessity of incorporating every parameter of interest into the ARG-inference algorithm. In contrast, importance sampling can be done without changing the basic inference algorithm and, therefore, might provide more flexibility for inferring a wider set of parameters. For example, one can imagine using ARGs to examine other processes, such as background selection, assortative mating, inference of admixture graphs, and others. We therefore consider the use of importance sampling of ARGs for full probabilistic inference of population genetic parameters to be an important line of future research for the population genetics community. Other important future directions in this area include the development of fast, properly calibrated ARG-inference methods and finding efficient ways to compute the probability of an ARG or coalescent tree under a given set of population genetic parameters.

4.3 Illustration on human data

To illustrate the utility of inferred ARGs in understanding population genetic processes, we applied SINGER to human sequencing data around the *ABO* and *MCM6* genes. Both are previously reported targets of natural selection, but we here show how to reproduce the signals using inferred ARGs.

The *ABO* gene, which determines an individual's blood type, is one of the most variable coding regions outside of the HLA region [86]. Comparative genomics analysis has shown evidence of trans-species polymorphism at this locus, and it has been hypothesized that balancing selection is maintaining its polymorphism in humans and other primates [86-88]. If this hypothesis is true, then we would expect to see an unusually ancient TMRCA in the ABO gene. In Figure 6A, we plotted the estimated TMRCA around the *ABO* locus in the CEU population from the 1000 Genomes Project [89]. The inferred TMRCA within this region occurred over 6 million years ago (Mya), roughly the speciation time between humans and chimpanzees [90]. These results are compatible with the hypothesis of balancing selection maintaining trans-species polymorphism at this locus. We also note that as SINGER uses a standard coalescence prior, it likely underestimates the most ancient coalescence times, meaning the true TMRCA at this locus may be even older than suggested here.

The derived allele at the rs4988235 SNP in the *MCM6* gene has been identified as a causal variant for lactase persistence in Europeans and is believed to have been positively selected after the introduction of dairy farming [91-94]. Here, we compare the overall branch-length-based diversity (as used in [39]) in all GBR samples from the 1000 Genomes Project [95] to the diversity in the haplotypes carrying the derived allele of rs4988235 (Figure 6B). The within-carrier diversity is substantially lower than the overall diversity, and the depletion of diversity among carriers spans a very long region. This is consistent with a strong, recent selective sweep as previously proposed [91-94]. The results presented here can be reproduced by code in the following GitHub repo: https://github.com/YunDeng98/ARG_review.git.

5 Conclusion

While full probabilistic population genetic inferences on genomic data might have seemed impossible a few years ago, new computational methods have facilitated the use of ARGs for addressing a number of problems in population genetics such as inferring ages of mutations, identifying fine-scaled relationships among individuals, detecting natural selection acting on traits and individual alleles, and much more. Although still in their infancy, methods using either MCMC or importance sampling on ARGs promise to provide probabilistic frameworks that can take advantage of the rich amount of information in full genomic data. Even without such methods, ARG inference provides a crucial tool for visualizing genetic variation and detailed genetic relationships. Not all methods for inferring ARGs perform equally well. There is generally a trade-off between accuracy and computational complexity. At the moment, SINGER appears to be the most accurate method for inferring ARGs in settings where it applies, but only methods such as ARG-Needle and tsinfer are applicable to GWAS size data sets. We expect numerous developments over the next few years that will improve on computational aspects of ARG inference and expand the applications of ARGs to fully move computational population genetics into the genomic era.

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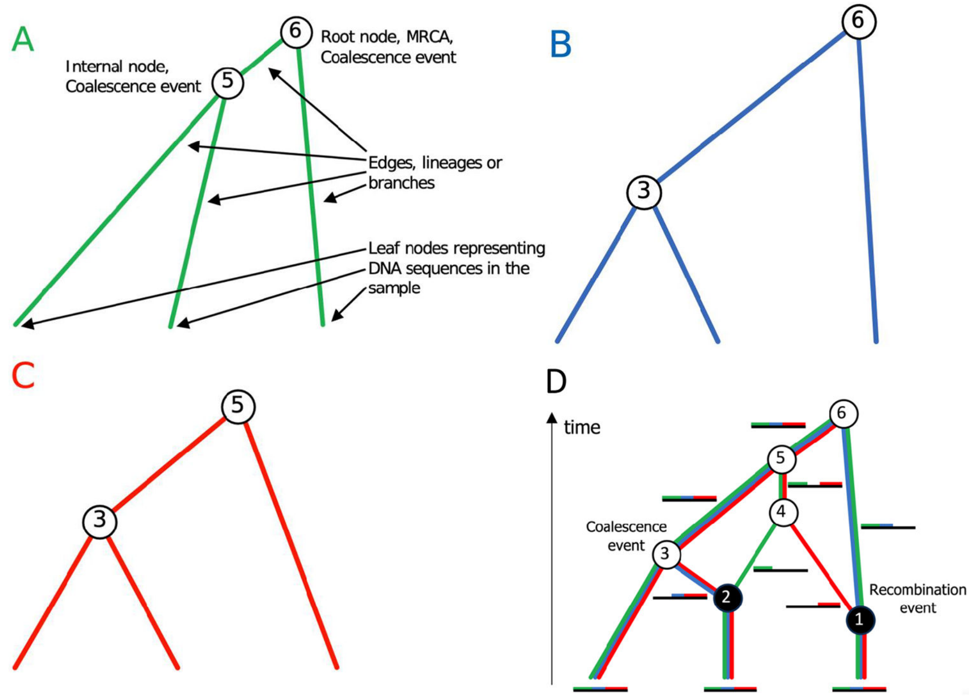

Fig. 1.

(A-C) Examples of coalescence trees. The trees consist of a set of nodes connected by edges. Synonymous words for edges used in biology are lineages or branches. The nodes representing the observed DNA sequences are the leaves (leaf nodes) in the tree, and the nodes representing coalescence events are internal nodes. The node at the top of the tree represents the Most Recent Common Ancestor (MRCA) of the three sequences. Only the internal nodes, and not the leaf nodes, are numbered and represented by circles in this depiction. Also, notice that the trees are depicted with the root at the top and the leaves at the bottom, as is the tradition in computer science (computer scientists do not spend much time in natural forests). All three coalescence trees (A, B, and C) are embedded in the ARG of (D). While the topologies are identical, the trees differ by having different branch lengths. (D) An example ARG. There are three sequences (bottom of figure). White circles represent coalescence events and black circles represent recombination events. The two recombination events split the sequences into three segments labeled with green, blue, and red colors.

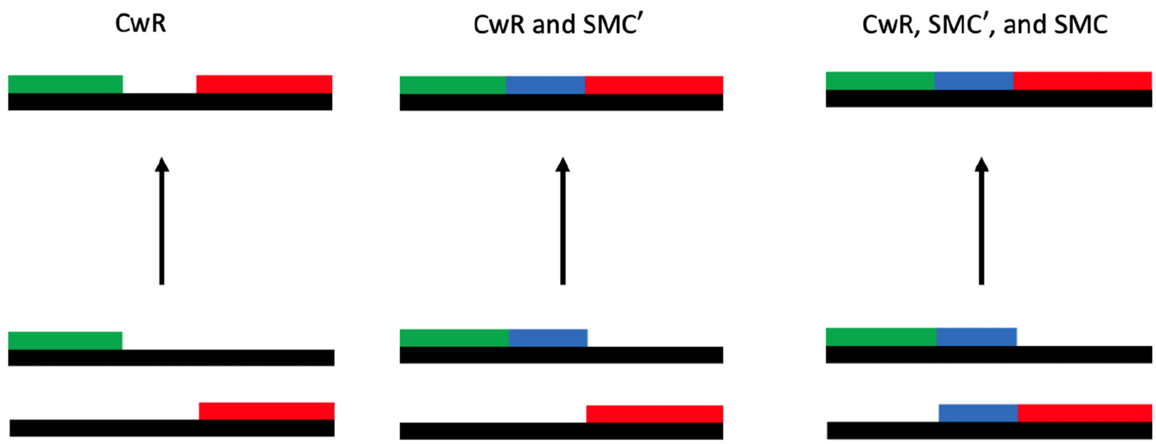

Fig. 2.

Examples of coalescence events permitted by different models of the coalescent process with recombination. The left panel illustrates a coalescence event between two lineages carrying non-overlapping and non-adjacent segments of DNA. Such coalescence events are allowed only under the full coalescent with recombination (CwR) and not by SMC′ or SMC. The middle panel illustrates a coalescence event between two lineages carrying adjacent but non-overlapping segments of DNA. Such coalescence events are allowed under both the CwR and SMC′ processes but not under SMC. The right panel illustrates a coalescence event between overlapping segments of DNA. Such coalescence events are allowed under all three models. The limits that each of the SMC′ and SMC models place on the types of allowed coalescences restrict the state space of ARGs that can be simulated under these models, with SMC inducing a more severe restriction than SMC′.

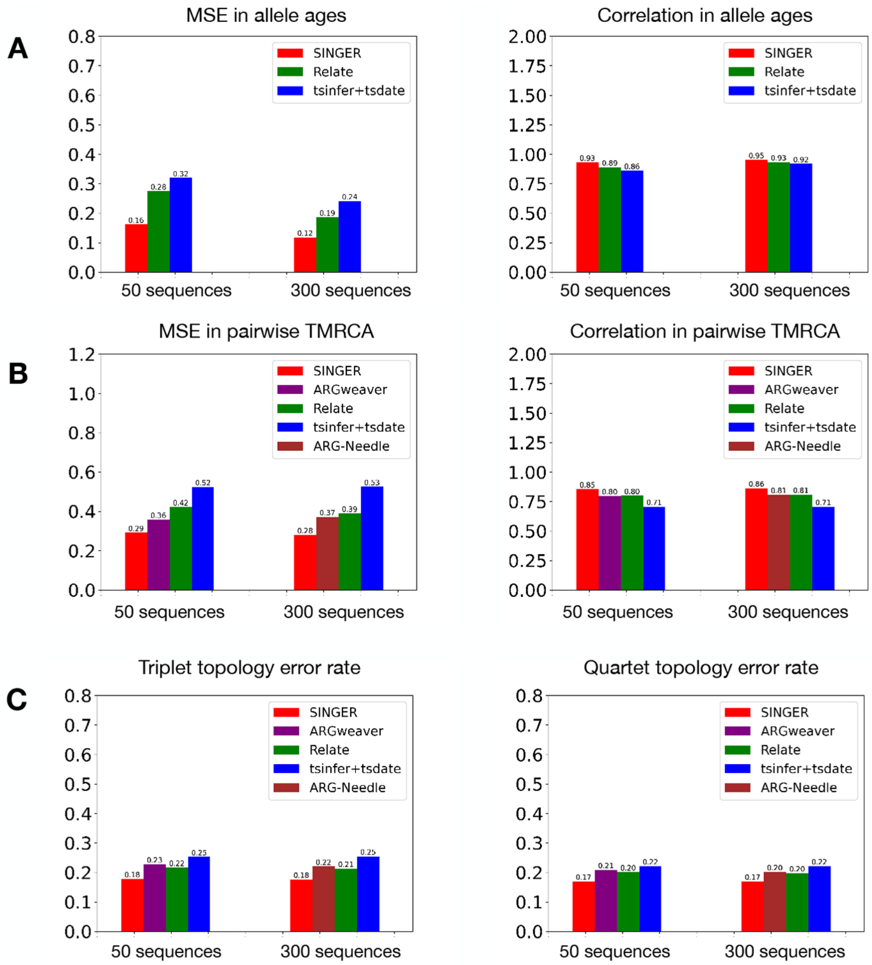

Fig. 3.

Performance of SINGER, ARGweaver, ARG-Needle, Relate and tsinfer+tsdate, evaluated with different metrics. These metrics include pairwise TMRCA (A), allele ages (B), triplet and quartet error rate (C). Methods were run on 50 and 300 simulated sequences. For allele ages and pairwise TMRCA, the mean squared error (MSE) and Spearman's correlation were calculated between the true values from the simulations and the inferred values. For topology benchmarking, we used the proportion of triplet or quartet topologies wrongly inferred (topology error rate) to measure performance.

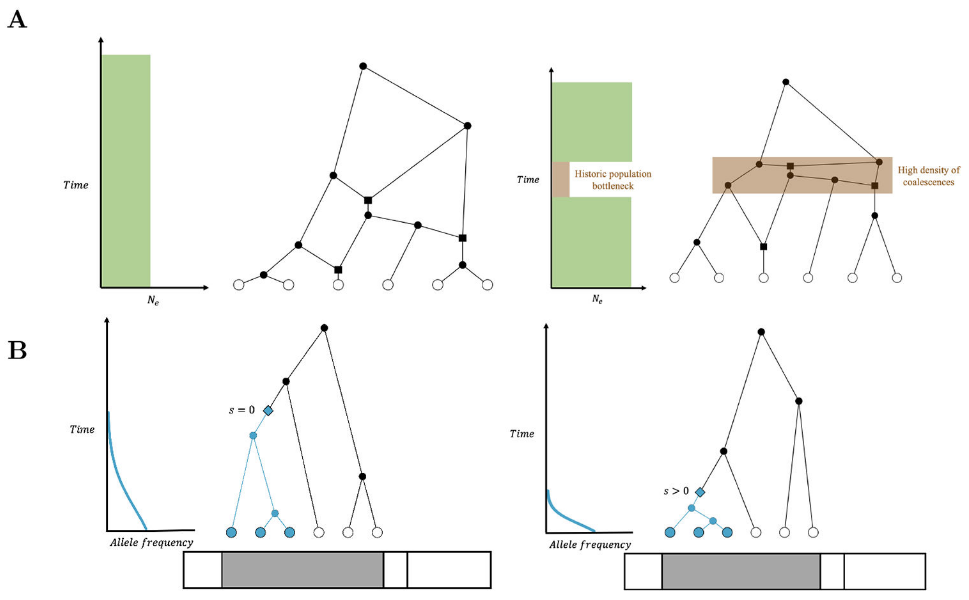

Fig. 4.

Examples of how information about different population genetic parameters are reflected in the ARG. (A) A smaller effective population size during a certain time period contributes to a higher density of coalescences across the whole ARG during that period. Note that here, as in Figure 5, we use squares to represent recombination events and circles for coalescence events. (B) Natural selection at a SNP results in a more rapid increase in the frequency of the selected allele, resulting in a higher density of coalescences of lineages carrying the selected allele (shown in blue) at the tree spanning the SNP. We highlight the portion of the genome this tree spans to emphasize that we are not showing the whole ARG.

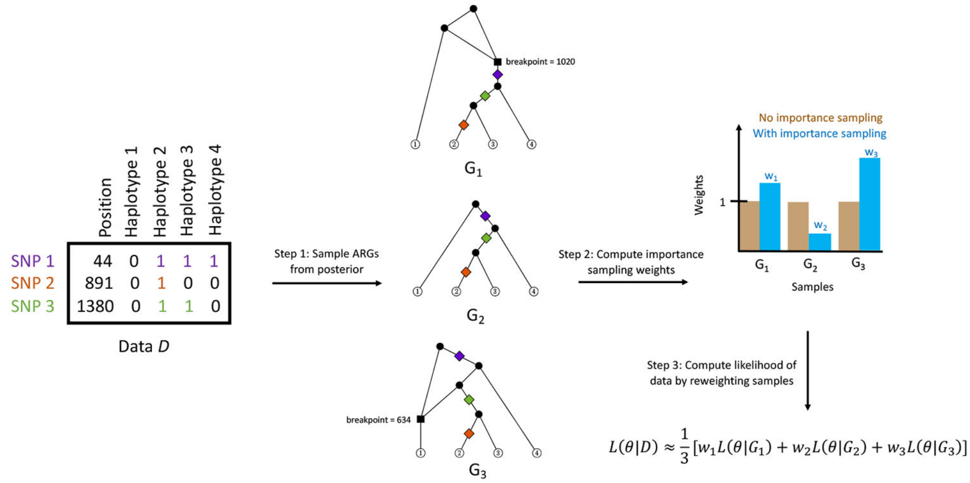

Fig. 5.

A schematic of how to utilize importance sampling with ARG inference methods to estimate the likelihood function of population genetic parameters. In Step 1, graphs are sampled from a certain posterior. As in Figure 4, mutations are shown as colored diamonds on the ARG, squares represent recombination events, and circles represent coalescence events. In Step 2, the importance sampling weight of each graph is computed. In Step 3, we compute the likelihood of a parameter as the weighted average of the likelihood of each sampled graph. Here θ represents any population genetic parameter of interest, such as population size or selection coefficient.

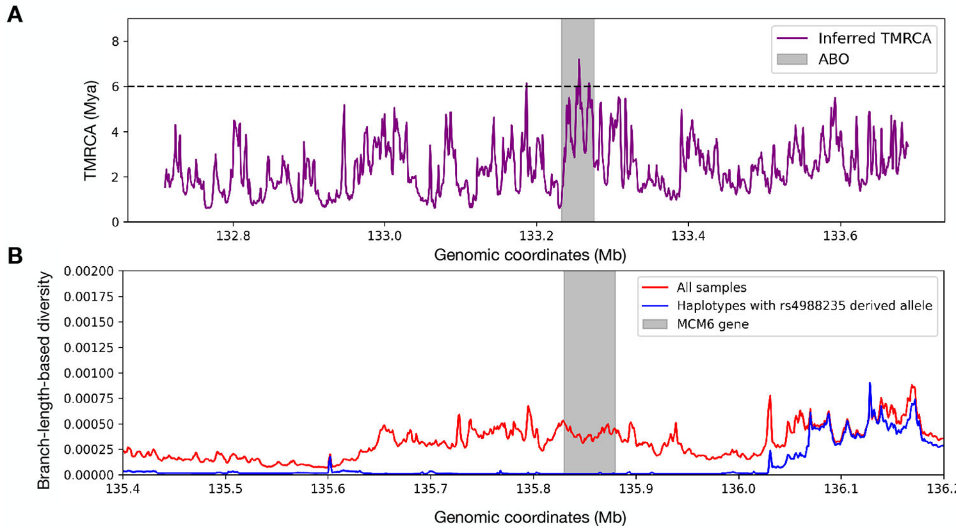

Fig. 6.

Using inferred ARGs to learn about the balancing selection at *ABO* and the selective sweep at *MCM6*. (A) The purple lines show the inferred TMRCA for 50 sequences from the CEU population from the 1000 Genomes Project, and the human-chimpanzee speciation time (around 6 Mya) is shown as a dashed line. The *ABO* gene (shaded area) exhibits coalescence times older than the speciation time. (B) The ARG-inferred 1 kb branch-length-based diversity among all samples (red) versus that of carriers of the derived allele of rs4988235 (blue) in the GBR population from the 1000 Genomes Project.

Table 1

The applicability of different ARG inference methods.

	Samples branch lengths	Samples topologies	Scales to 1000 Genomes Project	Scales to UK Biobank	Supports ancient DNA
ARGweaver-D	Yes	Yes	No	No	Yes
Relate	Yes	No	Yes	No	Yes
tsinfer+tsdate	No	No	Yes	Yes	Yes
ARG-Needle	No	No	Yes	Yes	No
SINGER	Yes	Yes	No	No	No